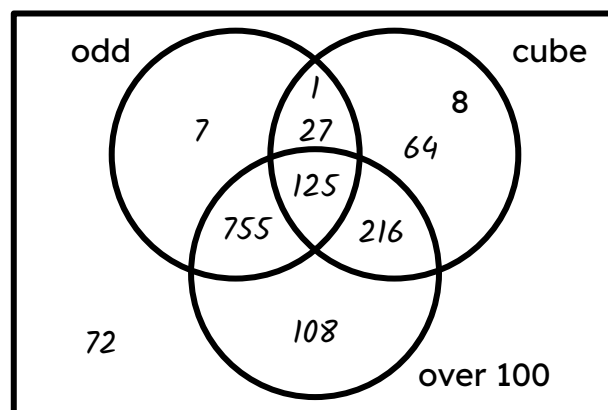
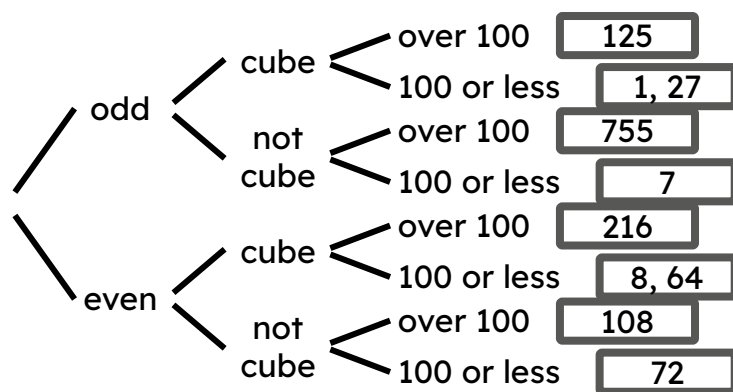




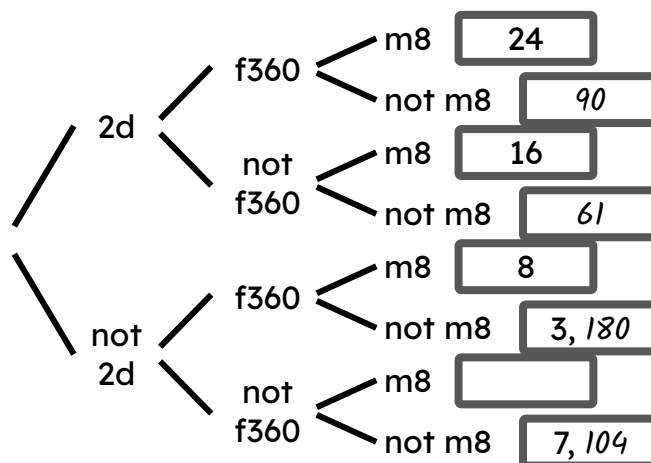
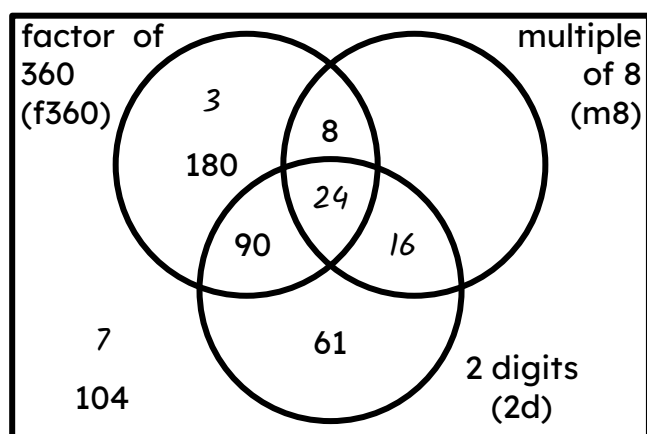
Comparing representations of outcomes for more than two events

Task A: Using different diagrams for multiple events

1) Using the correctly completed outcome tree, fill in the incomplete Venn diagram.



2) The Venn diagram and outcome tree both represent the same set of events for a spinner. Fill in all missing outcomes for both representations.

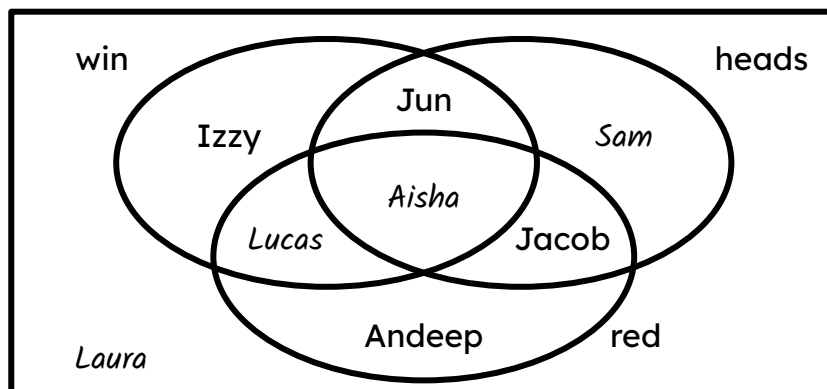


heads		
	win	lose
red	Aisha	Jacob
not red	Jun	Sam

tails		
	win	lose
red	Lucas	Andeep
not red	Izzy	Laura

3) Whilst one outcome table cannot show three events from a trial, two outcome tables can.

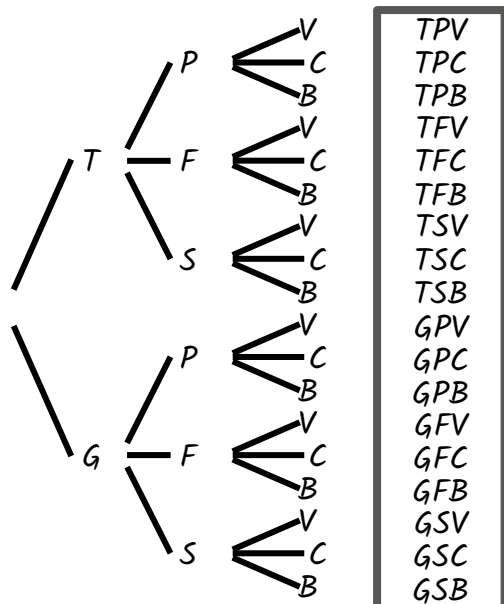
Fill in all missing outcomes on both the pair of outcome tables and the Venn diagram.



Task B: Multi-stage trials with lists and outcome trees

1) A three-course meal is made from one starter, one main, and one dessert.

By creating an outcome tree, list all possible three-course meals.



sample space

18 outcomes

Starters

Tomato Soup
Garlic Bread

Mains

Pasta Carbonara
Fish Pie
Spinach Gnocchi

Desserts

Vanilla Ice Cream
Chocolate Cake
Banana Split

2) A two-course meal is made from either:

one starter and one main
one main and one dessert

Create two lists of outcomes to find all possible two course meals.

starter
main

T	P
T	F
T	S
G	P
G	F
G	S

main
dessert

P	V
P	C
P	B
F	V
F	C
F	B
S	V
S	C
S	B

$6 + 9 = 15$ outcomes



Starters

Tomato Soup
Garlic Bread

Mains

Pasta Carbonara
Fish Pie
Spinach Gnocchi

Desserts

Vanilla Ice Cream
Chocolate Cake
Banana Split

3) How many more three-course meals are there than two-course meals?

There were 18 three-course meals, whilst only 15 two-course meals.

There were 3 more three-course meals.

4) Izzy has started to write down a list of outcomes for a three-stage trial.

Each outcome from every stage of the trial is represented at least once in Izzy’s incomplete list.

Draw and complete an outcome tree to create a sample space of all outcomes.

4 A T
4 R T
1 F T
7 R H
7 A H
7 K H

Here are the 24 outcomes.

There were:

three outcomes for the first stage of the event
four outcomes for the second stage of the event
two outcomes for the third stage of the event

